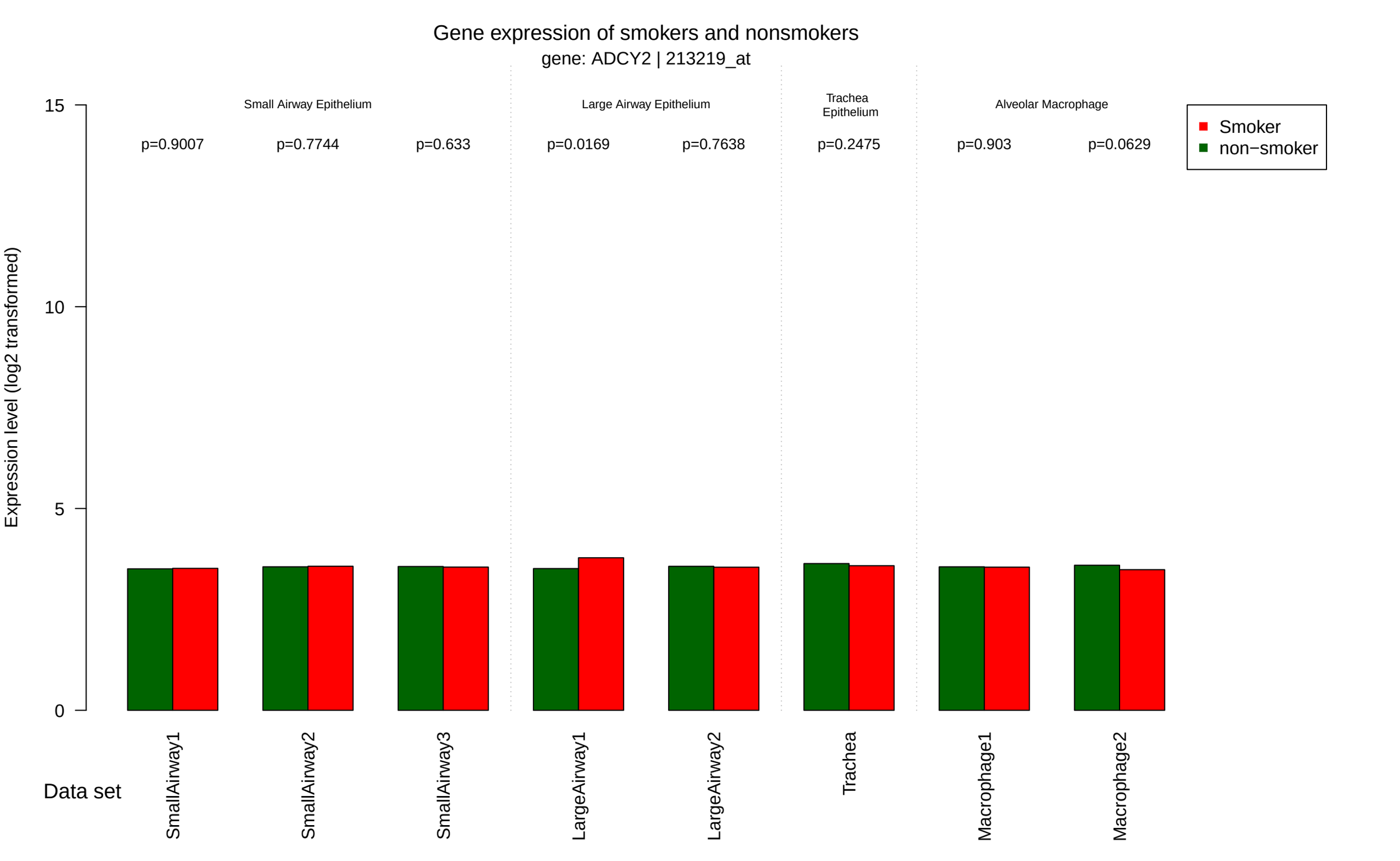
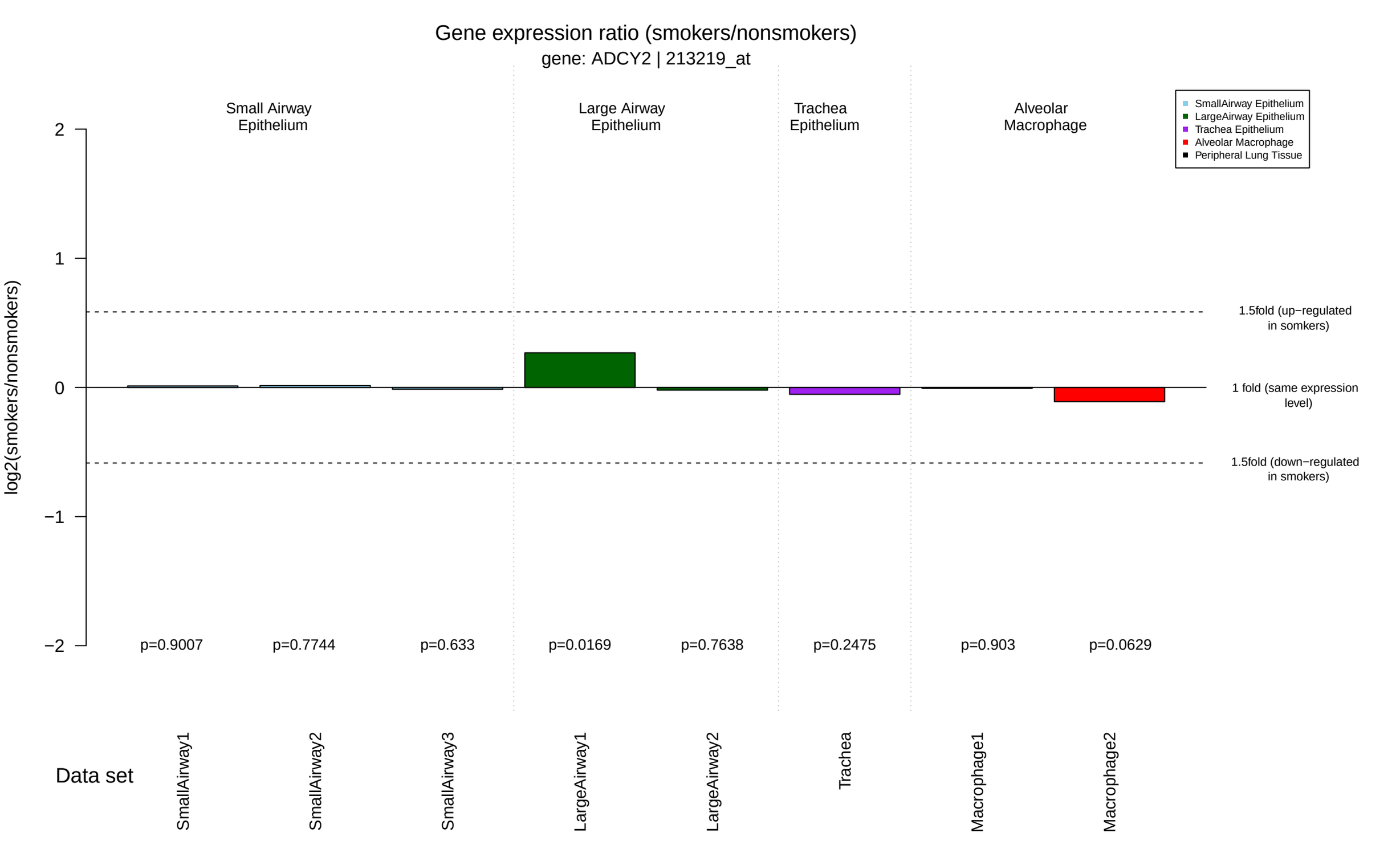
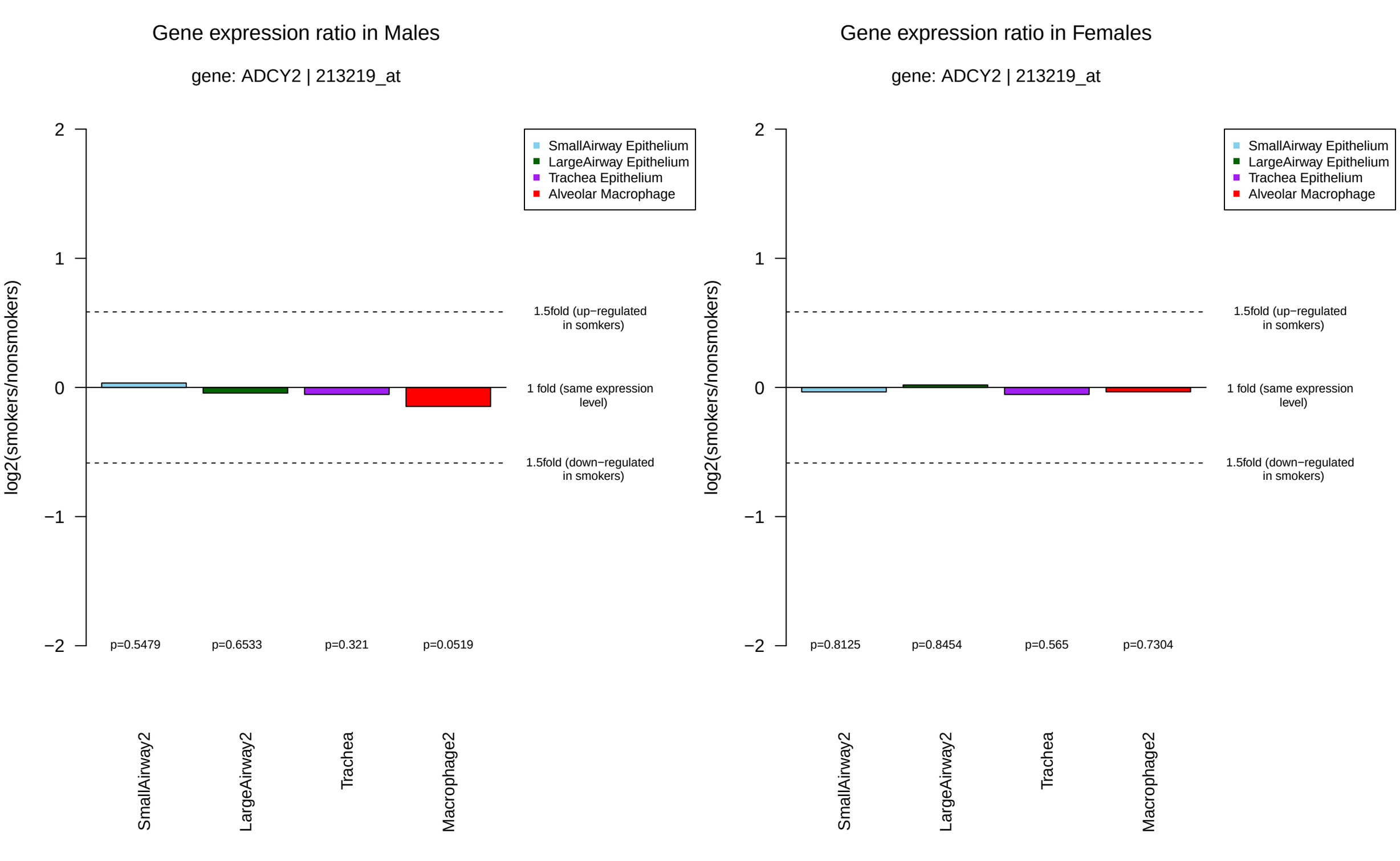
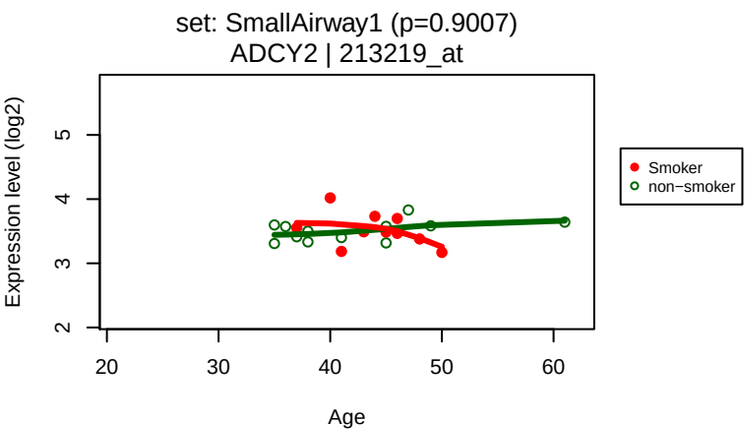
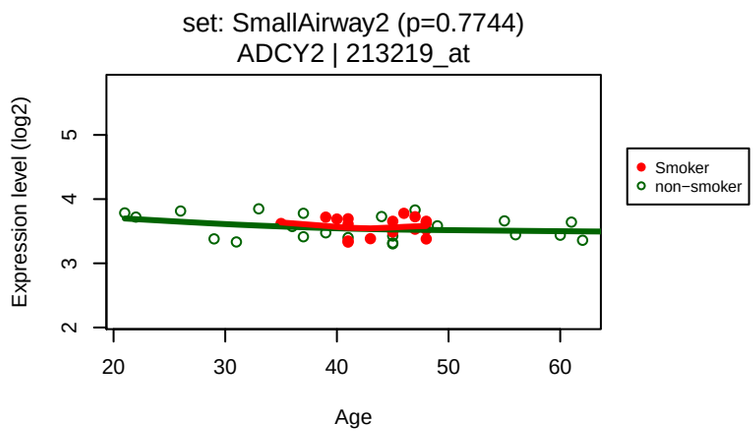
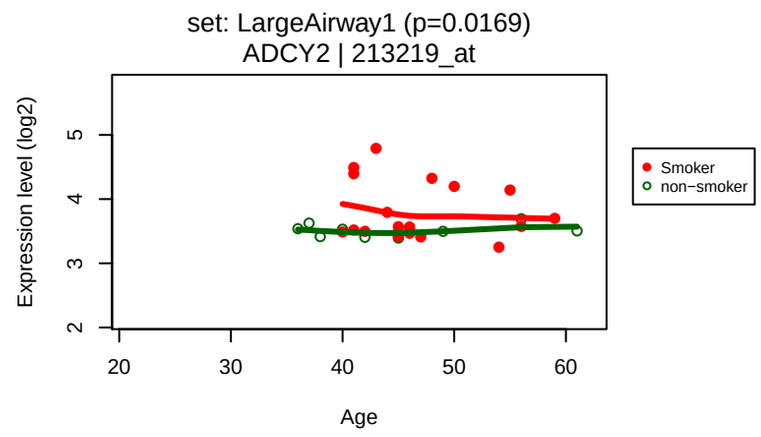

Gene expression of smokers and nonsmokers
gene: ADCY2 | 213219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9007
p=0.7744
p=0.633
p=0.0169
p=0.7638
p=0.2475
p=0.903
p=0.0629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

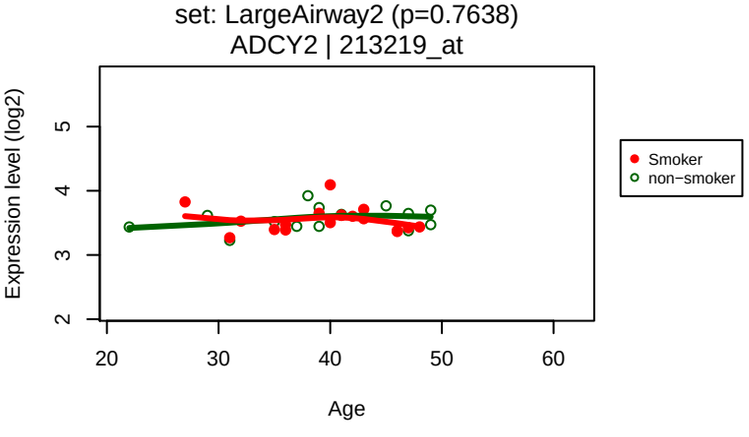

Gene expression ratio (smokers/nonsmokers)
gene: ADCY2 | 213219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9007
p=0.7744
p=0.633
p=0.0169
p=0.7638
p=0.2475
p=0.903
p=0.0629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

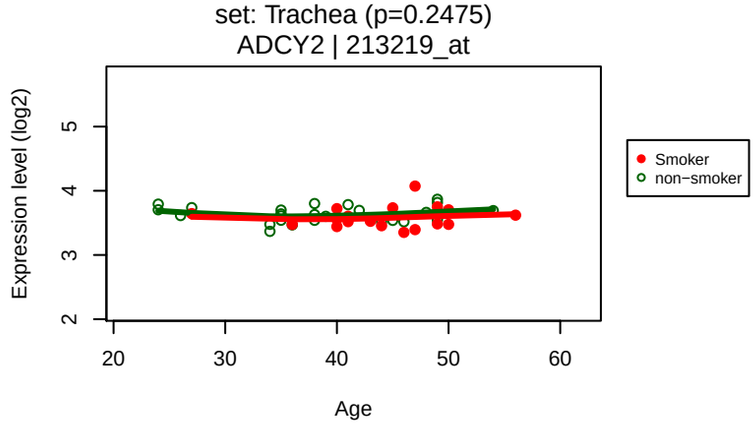

Gene expression ratio in Males
gene: ADCY2 | 213219_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5479
p=0.6533
p=0.321
p=0.0519
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY2 | 213219_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8125
p=0.8454
p=0.565
p=0.7304
SmallAirway2
LargeAirway2
Trachea
Macrophage2

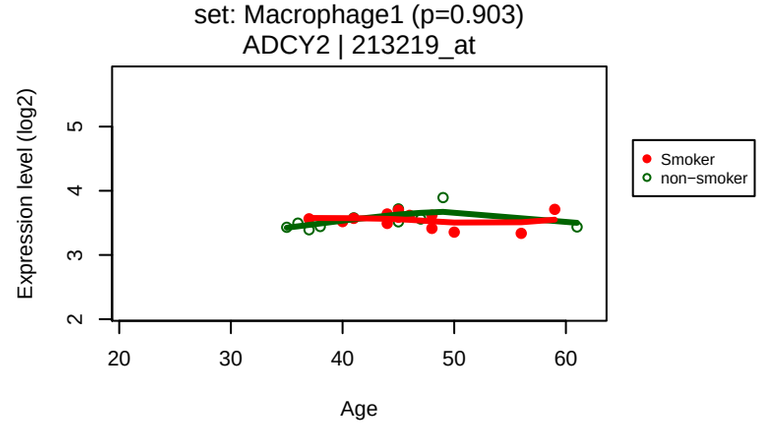

set: SmallAirway1 (p=0.9007)
ADCY2 | 213219_at
Expression level (log2)
Age
Smoker
non−smoker

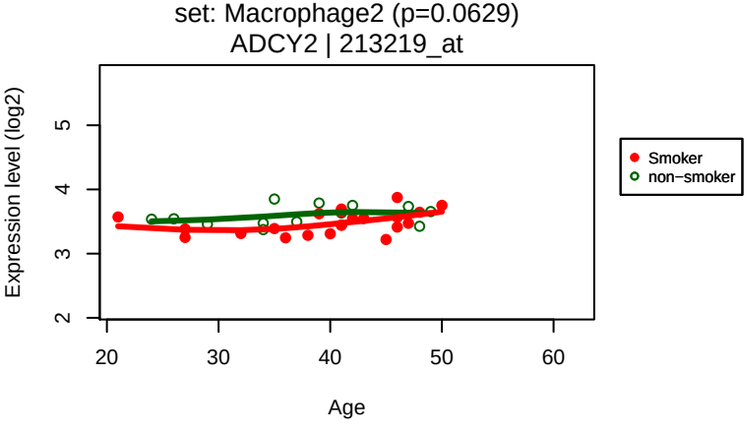

set: SmallAirway2 (p=0.7744)
ADCY2 | 213219_at
Expression level (log2)
Age
Smoker
non−smoker

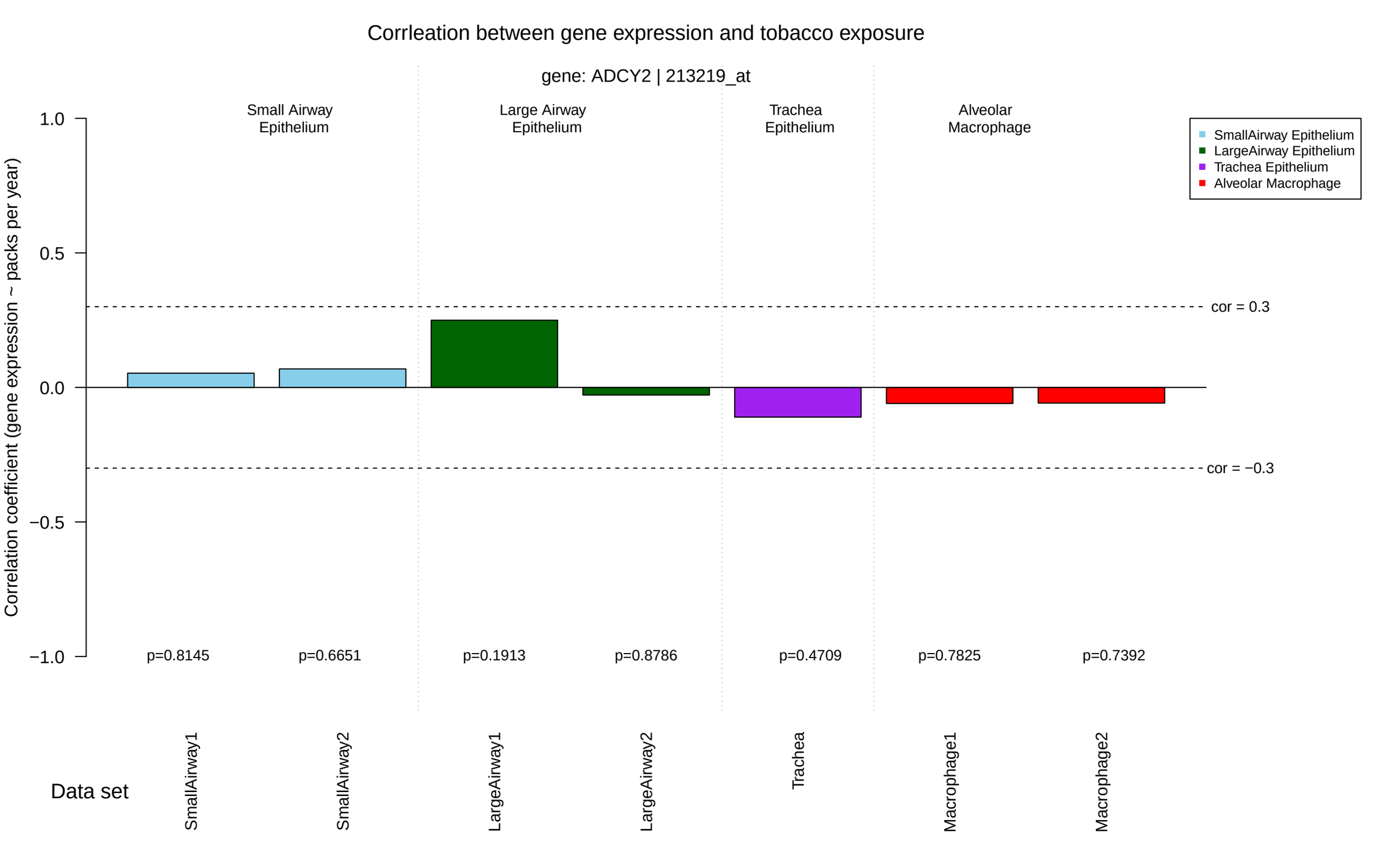

set: LargeAirway1 (p=0.0169)
ADCY2 | 213219_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7638)
ADCY2 | 213219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2475)
ADCY2 | 213219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.903)
ADCY2 | 213219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0629)
ADCY2 | 213219_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY2 | 213219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8145
p=0.6651
p=0.1913
p=0.8786
p=0.4709
p=0.7825
p=0.7392
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2